

2.1 Evolution

From the time we homo-sapiens learned to understand sound and create it, we have also played around with it. Be it anywhere on this planet, irrespective of religion and caste, all stories have had music associated with them. Fables usually had a bard singing for his damsel, or the Pied Piper luring the children with his captivating tunes. Music is everywhere and with music comes sound.

Although we know about Bach, Beethoven and Tansen, none of their original renditions are available to us as played by them. That's because there were no methods of recording anything at the time, except in the form of written musical notes.

Thomas Edison was someone who changed the scene with his invention of a device that could record and playback sounds. Christened the phonograph, this invention came about as a by-product of another experiment he was conducting to develop the Morse code. The technology that led to the phonograph came from developments that Edison made in the telegraph and telephone. At the time, Edison was experimenting with how a moving diaphragm that was linked to a coil could produce a voice-modulated signal. Meanwhile, he was also experimenting with a telegraph repeater that was simply a device that used a needle to indent paper with the dots and dashes of the Morse code.

From these two ideas came the concept of attaching the stylus from a telegraph repeater to the diaphragm in the mouthpiece of a telephone. The first test in July 1877 involved a sheet of paper pulled under the needle, mechanically coupled to a diaphragm, as he shouted into the mouthpiece. This however, didn't work but it did produce an unrecognisable sound, which was enough to prove that the concept was right.

Edison then went on to improvise on this and replaced the paper sheet with tin foil. The tin foil was mounted on a cylinder, and the cylinder was turned via a hand crank during recording